

Joshua S. North

Climate and Ecosystem Sciences
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PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory, Berkeley, California. (August 2022 - Present)

Career Track Research Scientist (April 2025 - Current)

Postdoctoral Fellow (August 2022 - April 2025)

Advisor: Dr. Mark D. Risser

EDUCATION

University of Missouri Columbia, Columbia, Missouri. (July, 2022)

Ph.D., STATISTICS

Advisors: Dr. Christopher K. Wikle and Dr. Erin M. Schliep

Dissertation: A Bayesian Approach to Data-Driven Discovery of Nonlinear Dynamic Equations

University of Missouri Columbia, Columbia, Missouri. (May, 2019)

M.A., STATISTICS

University of Colorado Boulder, Boulder, Colorado. (May, 2017)

B.S., APPLIED MATHEMATICS

B.A., ECOLOGY AND EVOLUTIONARY BIOLOGY

Minor, STATISTICS

PUBLICATIONS

Mahesh, A., Collins, W., Bonev, B., Brenowitz, N., Cohen, Y., Elms, J., Harrington, P., Kashinath, K., Kurth, T., **North, J. S.**, O'Brien, T., Pritchard, M., Pruitt, D., Risser, M., Subramanian, S., & Willard, J. (2025). "Huge ensembles part I: Design of ensemble weather forecasts using spherical fourier neural operators". *Geoscientific Model Development*, (17), 5575–5603. <https://doi.org/10.5194/gmd-18-5575-2025>. Accepted

Mahesh, A., Collins, W., Bonev, B., Cohen, Y., Harrington, P., Kashinath, K., Kurth, T., **North, J. S.**, O'Brien, T., Pritchard, M., Pruitt, D., Risser, M., Subramanian, S., & Willard, J. (2025). "Huge ensembles part II: Properties of a huge ensemble of hindcasts generated with spherical fourier neural operators". *Geoscientific Model Development*, (17), 5605–5633. <https://doi.org/10.5194/gmd-18-5605-2025>. Accepted

- Wikle, C. K., **North, J. S.**, Gopalan, G., & Yoo, M. (2025). “A statistician’s overview of mechanistic-informed modeling”. *arXiv*. <https://doi.org/2507.14336>. Under review in *Journal of the American Statistical Association*
- North, J. S.**, Wikle, C. K., & Schliep, E. M. (2025). “A Bayesian approach for spatio-temporal data-driven dynamic equation discovery”. *Bayesian Analysis*, (20), 375–404. <https://doi.org/10.1214/23-BA1406>
- Zhou, Y., North, J. S., Rhoades, A. M., Tao, J., Rudisill, W., Risser, M. D., & Collins, W. D. (2025). “Atmospheric river frequency-category characteristics shape u.s. west coast runoff”. *Journal of Geophysical Research: Atmospheres*, 130(2), e2024JD041805. <https://doi.org/https://doi.org/10.1029/2024JD041805>
- North, J. S.**, Risser, M. D., & Breidt, F. J. (2024). “A flexible class of priors for orthonormal matrices with basis function-specific structure”. *Spatial Statistics*, 64, 100866. <https://doi.org/10.1016/j.spasta.2024.100866>
- Custer, C. A., **North, J. S.**, Schliep, E. M., Verhoeven, M. R., Hansen, G. J. A., & Wagner, T. (2024). “Predicting responses to climate change using a joint species, spatially dependent physiologically guided abundance model”. *Ecology*, e4362. <https://doi.org/10.1002/ecy.4362>
- North, J. S.**, Schliep, E. M., Hansen, G. J. A., Kundel, H., Custer, C. A., McLaughlin, P., & Wagner, T. (2024). “Accounting for spatiotemporal sampling variation in joint species distribution models”. *Journal of Applied Ecology*, 1–16. <https://doi.org/10.1111/1365-2664.14547>
- North, J. S.**, Wikle, C. K., & Schliep, E. M. (2023). “A Review of Data-Driven Discovery for Dynamic Systems”. *International Statistical Review*, 91(3), 464–492. <https://doi.org/10.1111/insr.12554>. *Top 10 most cited ISR 2023 publication.*
- Wagner, T., Schliep, E. M., **North, J. S.**, Kundel, H., Custer, C. A., Ruzich, J. K., & Hansen, G. J. A. (2023). “Predicting climate change impacts on poikilotherms using physiologically guided species abundance models”. *Proceedings of the National Academy of Sciences*, 120(15), 1–8. <https://doi.org/10.1073/pnas.2214199120>
- North, J. S.**, Wikle, C. K., & Schliep, E. M. (2022). “A Bayesian approach for data-driven dynamic equation discovery”. *Journal of Agricultural, Biological, and Environmental Statistics*, 1(1), 1–28. <https://doi.org/10.1007/s13253-022-00514-1>. *ENVR student paper competition - honorable mention; JABES 2022 best paper award - honorable mention.*
- North, J. S.**, Schliep, E. M., & Wikle, C. K. (2020). “On the spatial and temporal shift in the archetypal seasonal temperature cycle as driven by annual and semi-annual harmonics”. *Environmetrics*, 1–16. <https://doi.org/10.1002/env.2665>
- North, J. S.**, Stanley, Z., Kleiber, W., Deierling, W., Gilleland, E., & Steiner, M. (2020). “A statistical approach to fast nowcasting of lightning potential fields”. *Advances in Statistical Climatology, Meteorology, and Oceanography*, 2, 79–90. <https://doi.org/10.5194/ascmo-6-79-2020>

SUBMITTED/PREPRINT

All preprints are available upon request.

Rhoades, A. M., **North, J. S.**, Rudisill, W., Hatchett, B. J., Risser, M., Beltran-Peña, A., Chen, X., Heggli, A., Hotaling, S., Huning, L., LaPlante, M., Mahesh, A., Marshall, A., McCrary, R., McEvoy, D., McGinnis, S., Rahimi, S., Randall, C., Srivastava, A., ...

- Jones, A. D. (2025). “Snow-eater heatwaves of the western united states”.
<https://doi.org/https://doi.org/10.21203/rs.3.rs-7576317/v1>. Submitted.
- North, J. S.**, Risser, M. D., & Rhoades, A. M. (2025). “Bayesian regression for dependent tensor-valued data from exponential families”. Under review in the *Journal of Agricultural, Biological, and Environmental Statistics*.
- Custer, C. A., **North, J. S.**, Schliep, E. M., Verhoeven, M. R., Link, D., Hansen, G. J. A., & Wagner, T. (2025). “Climate-driven declines in abundance across thermal guilds in fish communities of 11,000 temperate lakes”. Under review in *Diversity and Distributions*.

PRESENTATIONS

* - Invited, † - Virtual, ‡ - Poster

Structured Orthonormal Matrices for Spatio-Temporal Statistical Models, University of New South Wales Sydney, Sydney, AU; July, 24, 2025.*†

A probabilistic model based approach for matrix factorization of extreme data, Contemporary Advances in Statistics of Extremes workshop, University of Missouri Columbia; July 1, 2025.*

A Bayesian Approach for Data-Driven Dynamic Equation Discovery, International Biometric Society Journal Club, JABES Best Paper Session; March 20, 2025.*†

Structured Orthonormal Matrices for Spatio-Temporal Statistical Models, Colorado School of Mines, Golden, CO; February 11, 2025.*

Methods for quantifying internal variability from observations, American Geophysical Union 2024 (AGU24) Washington D.C.; December 13, 2024.‡

Quantifying internal variability from observations: A case study for the Pacific Northwest, 2024 ENVR Workshop: Spatial Data Science for the Environment, NCAR, Boulder, CO, October 4, 2024.‡

Quantifying internal variability from observations: A case study for the Pacific Northwest, Regional and Global Model Analysis monthly meeting, October 3, 2024.*†

Uncertainty quantification for low-likelihood high-impact weather events using spatio-temporal statistical modeling, Joint Statistical Meetings, Portland, Oregon, August 6, 2024

Data-driven dynamical modeling for spatio-temporal statistics,

- University of Wyoming, Laramie, WY; February 19, 2024.*
- Cornell University, Ithaca, NY; February 7, 2024.*
- University of California Santa Cruz, Santa Cruz, CA; January 22, 2024.*
- University of Virginia, Charlottesville, VA; January 17, 2024.*
- Indiana University Bloomington, Bloomington, IN; December 14, 2023.*
- Wake Forest University, Winston-Salem, NC; December 7, 2023.*

A statistical model for structured empirical orthogonal functions, American Geophysical Union 2023 (AGU23), San Francisco, CA; December 11, 2023.

A flexible prior for structured orthonormal matrices,

- South Dakota State University, Brookings, SD; September 29, 2023.*[†]
- Joint Statistical Meetings 2023, Toronto, Canada; August 8, 2023.
- Spatial Statistics 2023 - Climate and the Environment, University of Colorado Boulder; July 21, 2023.

A flexible class of priors for conducting posterior inference on structured orthonormal matrices, Climate Extremes Workshop, Clemson University; May 16, 2023. [‡]

A flexible class of priors for conducting posterior inference on structured orthonormal matrices, Machine Learning and Analytics Group, Lawrence Berkeley National Laboratory; May 11, 2023.*

A Bayesian Approach for Spatio-Temporal Data-Driven Dynamic Equation Discovery, ENVR 2022 Workshop, Environmental and Ecological Statistical Research and Applications with Societal Impacts, Provo, UT; October 6, 2022. [‡]

A Bayesian Approach for Data-Driven Dynamic Equation Discovery, Joint Statistical Meeting, Washington D.C.; August 10, 2022.

A Bayesian Approach to Data-Driven Discovery of Nonlinear Dynamic Equations, Lawrence Berkeley National Laboratory, Berkeley, CA; January 21, 2022.*[†]

A Bayesian Approach to Data-Driven Discovery of Nonlinear Dynamic Equations, Sandia National Laboratory, Albuquerque, NM; December 14, 2021.*[†]

Data-Driven Approach to Nonlinear Dynamic Equation Discovery, Joint Statistical Meeting; August 9, 2021.[†]

On the Spatial and Temporal Shift in the Archetypal Seasonal Temperature Cycle as Driven by Annual and Semi-Annual Harmonics, Joint Statistical Meeting; Virtual, August 4, 2020. *[†]

Accuracy of Radial Support Vector Classifiers; Effect of Imbalanced Training Sets on Varying Minority Class Prevalence, University of Colorado Boulder, Boulder, Colorado; May 1, 2017. [‡]

Creating Reproducible Research, University of Colorado Boulder, Boulder, Colorado; March 21, 2017.

FUNDING

\$3,000,000/yr - Senior Personnel, DOE Office of Science, Calibrated and Systematic Characterization, Attribution, and Detection of Extremes (CASCADE) Science Focus Area (SFA). <https://cascade.lbl.gov/>. (2024 - 2027)

Data-driven methods and open-source software toolkit for analyzing compound extremes. White paper, decision pending.

Scientific Machine Learning for Complex Systems, DE-FOA-0002958. Not awarded.

TEACHING

University of Missouri Columbia, Columbia, Missouri.

Course Development

STAT 4330/7330 - Methods in Sports Analytics I

STAT 4340/7340 - Methods in Sports Analytics II

Graduate Teaching Assistant

STAT 4340/7340 - Methods in Sports Analytics II (S20, S21, S22)

STAT 4330/7330 - Methods in Sports Analytics I (F20, F21)

STAT 3500 - Introduction to Probability and Statistics II (F18)

STAT 2500 - Introduction to Probability and Statistics I (F17, S18)

University of Colorado Boulder, Boulder, Colorado.

Undergraduate Teaching Assistant

APPM 3570 - Applied Probability (S16)

APPM 1235 - Pre-Calculus for Engineers (F15)

ADVISING

Rae Fadlovich (Summer 2025) - Department of Energy Computational Science Graduate Fellow

Matthew Murnane (Summer 2025) - DOE-RENEW CREW-RA, co-advisor with Alan Rhoades

Robert Born (Summer 2024) - DOE-RENEW CREW-RA, co-advisor with Alan Rhoades

Karthik Tadigiri (Summer 2024) - DOE-RENEW CREW-RA, co-advisor with Alan Rhoades

AWARDS

Top Cited Article: Wiley. Top 10 most cited 2023 International Statistical Review publication.

Journal of Agricultural, Biological, and Environmental Statistics 2022 best paper award -
Honorable Mention

ENVR Student Paper competition - Honorable Mention (2022)

PROFESSIONAL ROLES

American Statistical Association Statistics and the Environment Section (ENVR), Publications
Chair Elect (Appointment begins January 1 2026)

American Statistical Association Advisory Climate Change Policy Committee Member (2025-2027)

Associate Editor for Advances in Statistical Climatology, Meteorology and Oceanography

PROFESSIONAL ACTIVITIES

Organizer for the session *Advanced Statistical Methods for the Earth Sciences: Innovations in Climatology, Meteorology, and Oceanography* at the American Geophysical Union (AGU) December, 2024

Organized the session *Climatology Through the Lens of Dynamic Spatio-Temporal Processes* at the Joint Statistical Meetings (JSM) August, 2024

Organized the *2024 CASCADE retreat and workshop*, Berkeley, CA, July 23-25, 2024

Organized the *Statmospheric Seminar Series*, Earth and Environmental Sciences Area, Lawrence Berkeley National Laboratory, Fall, 2023

Graduate Student Leader, Space-Time Reading Group, University of Missouri (2021-2022)

Instructor and VIP Consultant for University of Missouri DataFest (2018-2022)

University of Missouri Statistics Graduate Student Association Vice-President (2017-2018)

University of Missouri Statistics Graduate Student Association Treasurer (2018-2019)

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA) (2017 - present)

International Society for Bayesian Analysis (ISBA) (2023 - present)

International Statistical Institute (ISI) (2025 - present)

The International Environmetrics Society (TIES) (2025 - present)

American Geophysical Union (AGU) (2023 - present)

REFeree

Bayesian Analysis; Communications in Statistics - Simulation and Computation; Earth System Dynamics; Engineering with Computers; Geoscientific Model Development; Geophysical Research Letters; Journal of Agricultural, Biological and Environmental Statistics; Journal of Climate; Journal of the American Statistical Association; Methods in Ecology and Evolution; Nonlinear Dynamics; Statistics and Public Policy; Spatial Statistics